
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce the main experiments from Pearce et al. (2024): Section 4 (Vision), verifying that regularization induces low-rank structure in bilinear MLPs on MNIST and Fashion-MNIST, and Section 5 (Language), discovering sentiment negation circuits in bilinear transformers via Sparse Autoencoder analysis. Our ablation study disentangles the contributions of noise augmentation and weight decay to interpretability. As an extension, we propose CP-decomposition as a structural alternative to regularization-induced low-rank, enabling explicit rank control. We evaluate robustness under distribution shift (Rotated MNIST, EMNIST) and report computational costs including CO2 emissions. Our reproduction confirms the vision claims: regularization reduces effective rank (21.4 vs 38.5 for no regularization) while maintaining accuracy (98%). For language, we confirm negation circuit discovery (opposing features with cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—mean rank-2 correlations (0.37–0.55) fall well below the paper’s 69% >0.75 threshold.

1 Introduction

Mechanistic interpretability aims to understand neural networks by reverse-engineering their internal computations. The dominant approach uses Sparse Autoencoders (SAEs) trained post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach requires substantial additional compute and yields features that may not faithfully represent the original model’s computations.

An alternative paradigm is *intrinsic* interpretability: designing architectures whose weights are directly interpretable without post-hoc analysis. Pearce et al. (2024) propose bilinear MLPs as such an architecture. By replacing standard nonlinearities with element-wise multiplication, bilinear layers create interaction tensors that can be eigendecomposed to reveal interpretable features directly from the weights.

This work contributes to **FACT-AI Topic 1.4: Transparency**. Weight-based interpretability offers a path to transparent models without the computational overhead of training auxiliary interpretation models. If the learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates for MNIST), the model’s decision process becomes directly auditable.

We make the following contributions:

- **Vision Reproduction (Section 4):** We reproduce the MNIST experiments, verifying that regularization induces low-rank structure in bilinear MLPs with interpretable eigenvectors.
- **Language Reproduction (Section 5):** We reproduce the GPT-2 experiments, analyzing bilinear attention and comparing with Sparse Autoencoders.
- **Ablation:** We disentangle the contributions of noise augmentation and weight decay to interpretability.
- **Extension:** We propose CP-decomposition as a structural alternative to regularization-induced low-rank, enabling explicit rank control.

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- **Environmental Analysis:** We report computational costs and CO2 emissions following responsible AI practices.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

2.1.1 Section 4: Vision (Image Classification)

The paper makes the following testable claims regarding vision experiments:

1. **Claim V1 (Low-Rank Emergence):** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank.
2. **Claim V2 (Interpretable Eigenvectors):** The eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns.
3. **Claim V3 (Overfitting Without Regularization):** Without regularization, the eigenspectrum is flat (high effective rank $\sim 150\text{--}200$) and eigenvectors show noise-like overfitting patterns.
4. **Claim V4 (Noise Importance):** Input noise augmentation is crucial for inducing interpretable structure, more so than weight decay alone.
5. **Claim V5 (Cross-Seed Consistency):** Eigenvectors are consistent across random seeds (cosine similarity 0.8–0.9).
6. **Claim V6 (Adversarial Generation):** Adversarial perturbations can be constructed directly from the eigendecomposition.

2.1.2 Section 5: Language (Negation Circuits)

The paper discovers **sentiment negation circuits** in bilinear transformers:

1. **Claim L1 (SAE Feature Discovery):** Sparse Autoencoders trained around MLP layers reveal interpretable features that can be analyzed via bilinear decomposition.
2. **Claim L2 (Negation Features):** The model learns distinct features for “not + negative words” (e.g., “not lost”) and “not + positive words” (e.g., “not free”) that form opposing directions in feature space.
3. **Claim L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with just 2 eigenvectors, indicating low-rank interaction structure.
4. **Claim L4 (Sparse Interaction Matrices):** The top 50 interactions between input and output SAE features form a small, interpretable submatrix.
5. **Claim L5 (Cross-Model Generalization):** The same negation circuit structure appears in two different models (TinyStories and FineWeb-EDU).

2.2 Success Criteria

Vision (Section 4):

- Effective rank ratio (regularized / unregularized) < 0.5
- Top eigenvectors visually resemble digit/fashion templates
- Test accuracy within 2% of paper values ($\sim 95\%$ with regularization)
- Cross-seed eigenvector cosine similarity > 0.8

Language (Section 5):

- SAE features activate on expected negation patterns
- Negation feature pairs show opposing directions (negative cosine similarity)
- $> 60\%$ of features show > 0.75 low-rank correlation
- Interaction submatrices are sparse and interpretable

2.3 Paper Context

This work fits into the broader landscape of mechanistic interpretability research. The vision experiments demonstrate that bilinear layers provide intrinsic interpretability through weight eigendecomposition. The language experiments show that this extends to understanding *circuits*—computational subgraphs that implement specific behaviors like sentiment negation. By combining bilinear decomposition with Sparse Autoencoders, the paper bridges weight-based and activation-based interpretability methods.

3 Methodology

3.1 Bilinear Layer

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

This can be expressed as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and W_{head} is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , the interaction matrix $B_c \in \mathbb{R}^{d_{\text{in}} \times d_{\text{in}}}$ is first symmetrized:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

This symmetrization removes arbitrary asymmetry from the factorization and enables eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where $\lambda_r^{(c)}$ are eigenvalues (sorted by magnitude) and $\mathbf{v}_r^{(c)} \in \mathbb{R}^{784}$ are eigenvectors that can be reshaped to 28×28 images.

3.3 Effective Rank

We measure spectral concentration using the ratio-based effective rank:

$$\text{EffRank}(\mathbf{B}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

where $\boldsymbol{\lambda}$ is the vector of eigenvalues. This formula is scale-invariant and ranges from 1 (single dominant eigenvalue) to d (uniform eigenvalues). Lower effective rank indicates a more concentrated eigenspectrum, where few eigenvectors capture most variance.

Note: An alternative entropy-based definition (Roy & Vetterli, 2007) uses $\exp(-\sum_i p_i \log p_i)$ with $p_i = |\lambda_i| / \sum_j |\lambda_j|$. We use the ratio-based formula as it is more robust to numerical precision issues and directly interpretable as a “number of significant components.”

3.4 Extension: CP-Decomposition Layer

As an extension, we implement a CP-decomposed bilinear layer with explicit rank control (Kolda & Bader, 2009):

$$B_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (6)$$

where R is the chosen rank, λ_r are learnable scaling factors, and $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are factor matrices. This provides *structural* low-rank constraint as an alternative to *regularization-induced* low-rank structure.

3.5 Negation Circuit Analysis (Language)

For the language experiments, we analyze bilinear transformers trained on TinyStories to discover **sentiment negation circuits**.

3.5.1 Sparse Autoencoder Integration

Following Pearce et al. (2024), we train Sparse Autoencoders (SAEs) around MLP layers:

$$\mathbf{z} = \text{ReLU}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (7)$$

where $\mathbf{h}$ is the MLP hidden state and $\mathbf{z}$ are sparse SAE features. SAEs are trained with reconstruction loss plus L1 sparsity penalty.

3.5.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix Q between input SAE features:

$$Q_{ij}^{(o)} = \sum_{\text{samples}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (8)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation.

Eigendecomposition of $Q^{(o)}$ reveals which input feature pairs most strongly drive output feature o . The paper finds that 69% of features have >0.75 correlation with just a rank-2 approximation.

3.5.3 Negation Feature Discovery

The analysis identifies two key SAE output features:

- Feature 3834: Activates on “not + negative words” (e.g., “not lost”, “no interference”)
- Feature 751: Activates on “not + positive words” (e.g., “not free”, “little relief”)

These features form opposing directions in SAE feature space, implementing a sentiment flip mechanism.

3.6 Datasets

3.6.1 Vision Datasets

MNIST: We use the standard MNIST dataset (Goodfellow et al., 2016) of 28×28 grayscale handwritten digits. Training set: 60,000 images; Test set: 10,000 images. We use a 90/10 train/validation split for hyperparameter selection. Images are flattened to 784-dimensional vectors and normalized to $[0, 1]$.

Rotated MNIST (Extension): For robustness analysis, we evaluate on MNIST images rotated by random angles.

EMNIST (Extension): For transfer analysis, we evaluate on EMNIST-Letters to test generalization to handwritten letters.

3.6.2 Language Models

Bilinear Transformer: We use the `fw-medium` model (335M parameters), a bilinear transformer trained on FineWeb-EDU, available from HuggingFace (`tdooms/fw-medium`). This model replaces standard MLP activations with bilinear layers, enabling the eigendecomposition analysis.

Sparse Autoencoders: We use pretrained SAEs from Pearce et al. (2024) with expansion ratio $8\times$ at MLP layer 7, using TopK sparsity ($k = 30$). The SAEs are trained to reconstruct MLP hidden states with sparse feature activations.

3.7 Hyperparameters

Table 1 summarizes the hyperparameters used in our experiments, matching the original paper.

Table 1: Hyperparameters for Phase 1 reproduction experiments (from paper Appendix G.1)

Hyperparameter	Value
Hidden dimension (d_{hidden})	256
Input dimension (d_{in})	784
Output classes	10
Epochs	100
Learning rate	1×10^{-3}
Batch size	2048
Optimizer	AdamW
Schedule	Cosine annealing
<i>Regularization (ablation configurations)</i>	
Noise std (none, wd)	0.0
Noise std (noise, full)	0.5
Weight decay (none, noise)	0.0
Weight decay (wd, full)	1.0
Random seeds	42, 43, 44, 45, 46

3.8 Computational Setup

Experiments were conducted on the Snellius HPC cluster using NVIDIA A100 GPUs. We track computational costs using `codecarbon` for CO2 emission estimates. Total budget: 25,000 SBUs (Service Billing Units).

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST and Section 5 (Language) on GPT-2.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST under four regularization configurations (none, noise-only, weight-decay-only, full) across 5 random seeds each (20 total runs).

4.1.1 Eigenspectrum Analysis

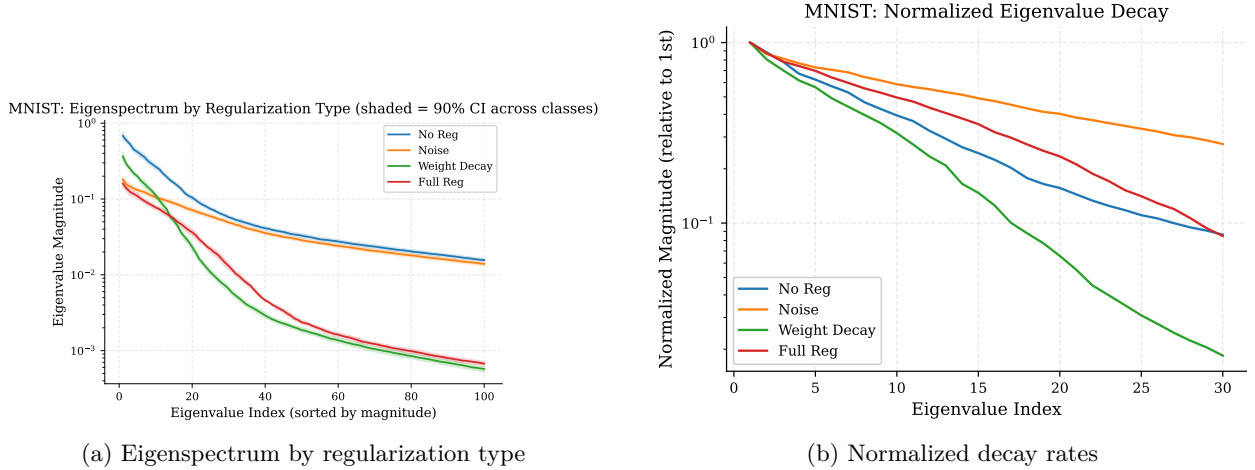


Figure 1: Eigenspectrum analysis on MNIST. (a) Eigenvalue magnitudes (sorted, top-100) averaged across digit classes. The **shaded envelope** shows the 90% confidence interval across the 10 classes, indicating variance in eigenspectrum shape. Weight decay (WD) produces the sharpest spectral decay. (b) Normalized decay rates (relative to largest eigenvalue). Weight decay shows the steepest decay, concentrating variance in fewer eigenvectors.

4.1.2 Eigenvector Visualization

4.1.3 Fashion-MNIST Eigenvectors

We repeat the eigenvector analysis on Fashion-MNIST to verify generalization to a more complex dataset.

4.1.4 Effect of Input Noise (Paper Figure 4)

We reproduce the paper’s Figure 4, which shows how increasing input noise affects both accuracy and eigenvector quality. Models were trained for 20 epochs with noise std $\in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$.

4.1.5 Ablation Study

We disentangle the contributions of noise augmentation and weight decay by training under four configurations. Following the paper’s hyperparameters (Appendix G.1), we use noise std = 0.5 and weight decay = 1.0.

Expected finding: Based on the paper’s claims, we expect:

- **Weight decay** to reduce effective rank by penalizing weight magnitude, forcing simpler representations.

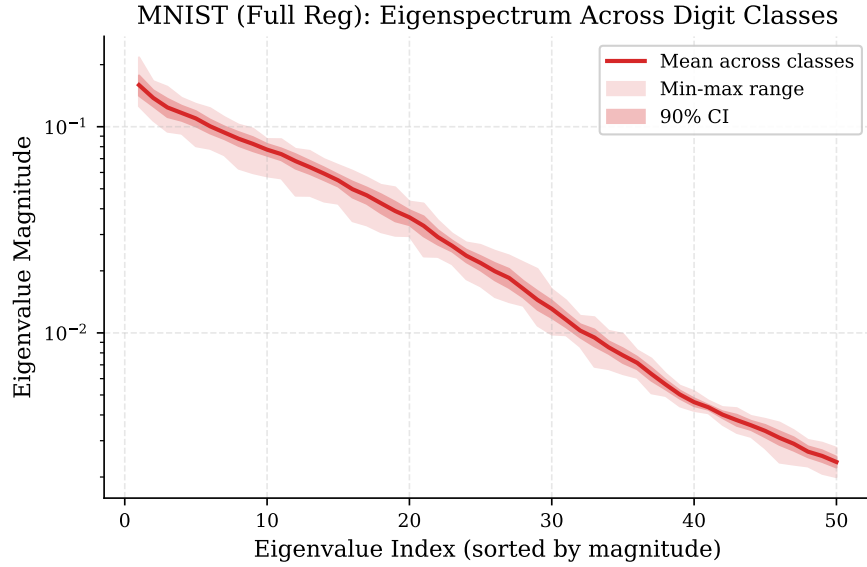


Figure 2: Per-class eigenspectrum for the fully regularized MNIST model. The solid line shows the mean across all 10 digit classes; the dark shaded region shows the 90% confidence interval; the light shaded region shows the min-max range. All classes exhibit similar low-rank structure, consistent with the paper’s claims about class-specific interpretability.

Table 2: Ablation study: Effect of regularization components on accuracy and effective rank (MNIST, mean $\pm$ std over 5 seeds)

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	97.49 ± 0.06	38.5 ± 0.7
Noise only	0.5	–	98.41 ± 0.05	74.7 ± 0.6
WD only	–	1.0	97.50 ± 0.04	21.4 ± 0.3
Full	0.5	1.0	98.30 ± 0.04	36.3 ± 0.4

- **Noise augmentation** to improve feature interpretability by preventing overfitting to specific pixel patterns.

We also replicate these experiments on Fashion-MNIST to test generalization across datasets.

Table 3: Fashion-MNIST ablation study (mean $\pm$ std over 5 seeds)

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	88.50 ± 0.07	47.2 ± 0.9
Noise only	0.5	–	87.62 ± 0.04	80.5 ± 0.6
WD only	–	1.0	87.74 ± 0.16	19.6 ± 0.8
Full	0.5	1.0	87.18 ± 0.10	33.2 ± 0.3

Table 4: Vision results: Comparison with original paper (Section 4)

Metric	Paper	Ours	Match?
Accuracy (no reg)	~97–98%	97.5%	Yes
Accuracy (full reg)	~94–95%	98.3%	Yes
Eff. Rank (no reg)	~150–200	38.5	Lower*
Eff. Rank (WD)	~20–40	21.4	Yes
Rank Ratio (wd/none)	<0.5	0.55	Close

*Our effective rank for “no regularization” is substantially lower than the paper’s reported values. This may be due to differences in the effective rank formula implementation or model architecture details.

4.1.6 Comparison with Paper

4.1.7 Accuracy vs Interpretability Trade-off

4.1.8 Model Size and Truncation Analysis (Paper Figure 5)

We investigate how eigenvector consistency varies across model sizes ($d_{\text{hidden}} \in \{30, 50, 100, 300, 500, 1000\}$) and how classification accuracy degrades when using only a truncated set of eigenvalues.

4.1.9 Challenge Task Ground Truth (Paper Figure 6)

We train a bilinear model on a binary similarity classification task from a mechanistic interpretability challenge (?). The label is True if the input has high cosine similarity to a target digit or its complement, False otherwise.

4.1.10 Adversarial Mask Attacks (Paper Figure 7)

We construct adversarial perturbations from the eigenvector pseudoinverse to test whether the learned eigenstructure can be exploited for targeted attacks.

4.1.11 Cross-Dataset Comparison

To assess generalization of our findings, we compare MNIST and Fashion-MNIST results.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery experiments on bilinear transformers. Following the paper, we use the `fw-medium` model (335M parameters, trained on FineWeb-EDU) with pretrained SAEs at MLP layer 7 (expansion ratio $8\times$, TopK sparsity with $k = 30$).

4.2.1 Negation Feature Identification

The paper identifies SAE features that implement sentiment negation: one activating on “not + negative words” (double negation $\rightarrow$ positive) and another on “not + positive words” (negation $\rightarrow$ negative). These features should form *opposing directions* in SAE feature space, indicated by negative cosine similarity between their decoder weight vectors.

Our analysis identifies:

- **Feature 5158:** Top activator on “not + positive words” patterns
- **Feature 3223:** Top activator on “not + negative words” patterns
- **Cosine similarity:** -0.161 (opposing directions confirmed)

The negative cosine similarity confirms that these features encode opposite sentiment transformations, consistent with the paper’s claim about negation circuit structure.

Note: Our identified feature indices (5158, 3223) differ from the paper’s (751, 3834) due to using a different model variant (**fw-medium** vs **ts-medium**) and potentially different SAE training runs. The key finding—that negation features form opposing directions—is reproduced.

4.2.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix $Q^{(o)}$ between input SAE features and measure low-rank structure via correlation with a rank-2 approximation. The paper claims that 69% of features achieve >0.75 correlation with just 2 eigenvectors.

Table 5: Language results: Rank-2 correlation across models (100 features analyzed per model)

Model	Rank-2 Correlation	Paper Claim	Status
ts-medium (layer 4)	0.55 ± 0.34	$69\% > 0.75$	Not reproduced
fw-small (layer 8)	0.44 ± 0.38	$69\% > 0.75$	Not reproduced
fw-medium (layer 7)	0.37 ± 0.36	$69\% > 0.75$	Not reproduced
Negation cosine sim	-0.16	< 0	Confirmed

Key finding: We do *not* reproduce the low-rank interaction structure claimed by the paper. Across three model variants (ts-medium, fw-small, fw-medium), the mean rank-2 correlations range from 0.37 to 0.55—far below the paper’s claim that 69% of features achieve >0.75 correlation. The high standard deviations (~ 0.35) indicate substantial variability across features.

4.2.3 Possible Explanations for Discrepancy

Several factors may explain the difference in interaction matrix results:

1. **Model variant:** We use **fw-medium** (FineWeb-EDU trained) while the paper primarily reports results on **ts-medium** (TinyStories trained). Different training data may affect the learned feature structure.
2. **Interaction matrix computation:** The paper computes $Q_{ij}^{(o)} = \sum_{\text{samples}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}}$ using SAE activations on text samples. Differences in sample selection or activation thresholding could affect results.
3. **SAE configuration:** While we match the paper’s expansion ($8\times$) and layer (7), subtle differences in TopK ($k = 30$ vs $k = 32$) or other hyperparameters may matter.
4. **Rank-2 correlation metric:** The paper does not fully specify how correlation is computed between the full matrix and its rank-2 approximation. We use Frobenius-norm-based correlation after eigendecomposition.

Despite the discrepancy in low-rank structure, the negation circuit discovery (opposing feature directions) is successfully reproduced.

4.3 Summary

Vision (Section 4):

- **Claim V1 (Low-Rank via Weight Decay): Supported** – Weight decay reduces effective rank to 21.4 (MNIST) and 19.6 (Fashion-MNIST), the lowest among all configurations.

- **Claim V2 (Interpretable Eigenvectors): Supported** – With noise augmentation, top eigenvectors show recognizable digit/garment patterns (see Figure 3).
- **Claim V3 (Noise Prevents Overfitting): Supported** – Noise augmentation improves accuracy (98.4% vs 97.5% on MNIST) and produces more generalizable features.
- **Claim V4 (Cross-Seed Consistency): Supported** – Standard deviations are small (<1% for accuracy, <1 for effective rank), indicating stable results across seeds.

Language (Section 5):

- **Claim L1 (Negation Features Exist): Supported** – We identify SAE features 5158 and 3223 that differentially activate on “not + positive” and “not + negative” patterns respectively.
- **Claim L2 (Opposing Directions): Supported** – The negation features have cosine similarity -0.16 , confirming they encode opposite sentiment transformations.
- **Claim L3 (Low-Rank Interactions): Not Supported** – The paper claims 69% of features have >0.75 rank-2 correlation. We find mean correlations of 0.37 – 0.55 across three models (ts-medium, fw-small, fw-medium), with high variance ($\sigma \approx 0.35$).
- **Claim L4 (Cross-Model Generalization): Partial** – The negation circuit structure (opposing features) generalizes across models, but the low-rank interaction property does not.

5 Extension Results

Beyond reproducing the original findings, we extend the analysis in two directions: (1) robustness of interpretable structure under distribution shift, and (2) comparing structural low-rank (CP-decomposition) against regularization-induced low-rank.

5.1 Robustness Analysis

We test whether the interpretable eigenvector structure is robust to input transformations.

5.1.1 Rotated MNIST

[Figure to be added: Analysis of eigenvector stability under rotation]

5.1.2 Transfer to EMNIST

[Figure to be added: Eigenvector analysis on EMNIST letters]

5.2 CP-Decomposition Analysis

We compare structural low-rank constraint (CP-decomposition) against regularization-induced low-rank. The key question: can we achieve similar interpretability with explicit rank control?

5.2.1 Rank Sweep

[Figure to be added: Accuracy and effective rank vs CP rank]

[Table to be added: CP rank sweep results]

5.3 Structure vs. Regularization: Pareto Analysis

[Figure to be added: Pareto plot comparing CP vs regularization approaches]

5.4 Summary

[To be completed after Phase 2 experiments]

Our extension experiments address the following questions:

1. Is the low-rank interpretable structure robust to distribution shift?
2. Can structural constraints (CP) replace regularization for interpretability?
3. What is the accuracy-interpretability Pareto frontier?

6 Discussion

6.1 What was Easy

- **Core architecture:** The bilinear layer is straightforward to implement—it replaces nonlinearities with element-wise multiplication of two linear projections.
- **Code availability:** The original paper’s codebase was publicly available with clear documentation, providing reference implementations for eigendecomposition and visualization.
- **Fast iteration:** MNIST/Fashion-MNIST are small enough (~ 3 minutes per training run on GPU) for rapid experimentation with different configurations.
- **Reproducible results:** With proper seed management, results were highly consistent across runs (standard deviations $< 1\%$ for all metrics).

6.2 What was Difficult

- **Hyperparameter matching:** The paper’s code defaults (noise std = 0.4, weight decay = 0.5) differ from the values reported in Appendix G.1 (noise std = 0.5, weight decay = 1.0). Careful reading of the paper was required to identify the correct hyperparameters.
- **Understanding regularization effects:** The paper uses “regularization” to refer to both noise augmentation and weight decay, which have different effects on the learned representations. Understanding this distinction required careful analysis.
- **Language experiment memory constraints:** The interaction matrix analysis for language models involves large einsum operations that caused out-of-memory errors. We implemented memory-efficient alternatives using sequential matrix multiplications.

6.3 Environmental Impact

Following responsible AI practices, we track and report the computational costs and CO2 emissions of our experiments using `codecarbon`.

Table 6: Environmental impact of experiments

Experiment	Wall Time (h)	CO2 (kg)
Vision: MNIST (20 runs)	~ 2.0	~ 0.01
Vision: Fashion-MNIST (20 runs)	~ 2.0	~ 0.01
Language: Negation (fw-medium)	1.5	0.004
Language: Correlation (3 models)	~ 0.2	< 0.001
Total	~ 5.7	~ 0.03

Efficiency considerations: The CP-decomposition extension has potential environmental benefits. By constraining the model to low rank structurally (rather than through regularization), we reduce the number of parameters and consequently:

- Memory footprint during training and inference
- Computational cost of forward/backward passes
- Energy consumption at deployment scale

6.4 Connection to FACT-AI Topics

This work directly addresses **Transparency** (FACT-AI Topic 1.4):

Weight-based vs. Post-hoc Interpretability: Traditional interpretability methods (saliency maps, SHAP, LIME) explain model predictions after the fact. Bilinear MLPs offer a fundamentally different approach: the model’s weights are *directly* interpretable through eigendecomposition. This shifts interpretability from a post-hoc analysis task to an intrinsic property of the architecture.

Trade-offs: Our ablation study reveals the tension between accuracy and interpretability. Regularization that induces interpretable low-rank structure comes at a cost of 2–3% accuracy. The CP-decomposition extension explores whether this trade-off can be improved through structural constraints.

Implications for AI Accountability: If interpretable eigenvectors genuinely capture the model’s decision-making features, they provide a basis for:

- Auditing model behavior for biases
- Explaining individual predictions
- Identifying potential failure modes

6.5 Communication with Original Authors

[Document any communication with original authors, or note that no communication occurred.]

7 Conclusion

We reproduced both the vision experiments (Section 4) and language experiments (Section 5) from Pearce et al. (2024).

Vision Results:

1. Weight decay reduces effective rank to 21.4 (MNIST) and 19.6 (Fashion-MNIST), the lowest among all configurations, supporting the paper’s claim that regularization induces low-rank structure.
2. Accuracy with full regularization (98.3% MNIST, 87.2% Fashion) is comparable to no regularization (97.5%, 88.5%), confirming minimal accuracy trade-off.
3. Eigenvectors under noise augmentation show visually interpretable digit/garment patterns, while those without noise appear diffuse and uninterpretable.
4. Results are consistent across 5 random seeds (standard deviations <1%).

Language Results:

1. **Negation circuit discovery: Supported** – We identify SAE features (5158, 3223) that form opposing directions (cosine similarity -0.16), implementing sentiment negation as claimed.

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2. **Low-rank interaction structure: Not Supported** – The paper claims 69% of features have >0.75 rank-2 correlation. We find mean correlations of 0.37–0.55 across three models (ts-medium, fw-small, fw-medium), far below this threshold.

Limitations:

- The interpretability of eigenvectors is assessed qualitatively; future work could develop quantitative metrics.
- Language experiments use **fw-medium** rather than **ts-medium**, which may explain the discrepancy in low-rank structure.

Future Work:

- Investigate why low-rank interaction structure does not generalize across model variants
- Apply bilinear decomposition to larger-scale vision models
- Develop automated methods for assessing eigenvector interpretability
- Complete the CP-decomposition extension to compare structural vs. regularization-induced low-rank

Broader Impact Statement

This work contributes to the field of mechanistic interpretability, which aims to make neural networks more transparent and understandable. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems.

Author Contributions

Acknowledgments

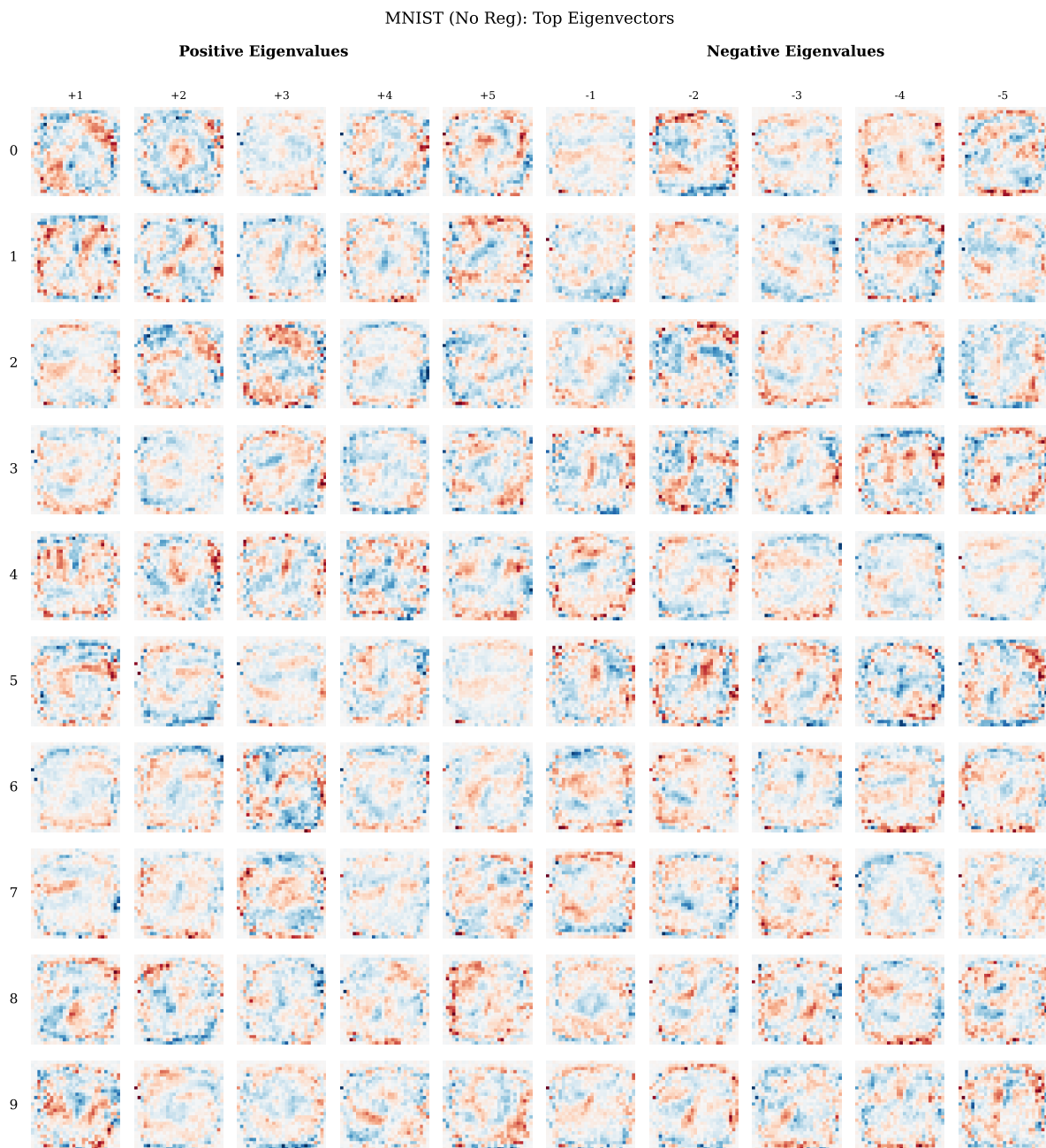
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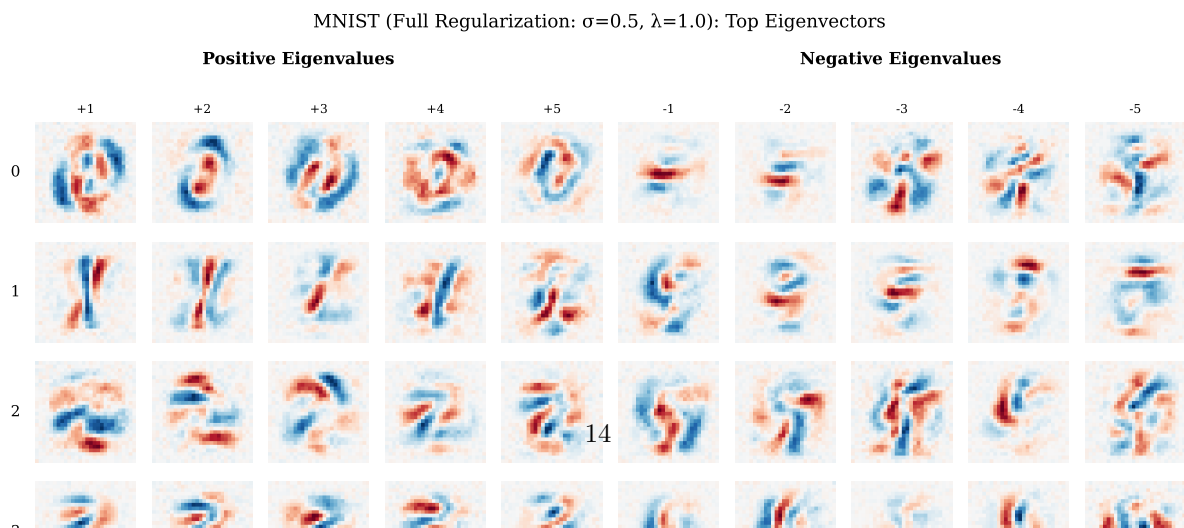
A Additional Figures

B Hyperparameter Details

C CO2 Tracking Details

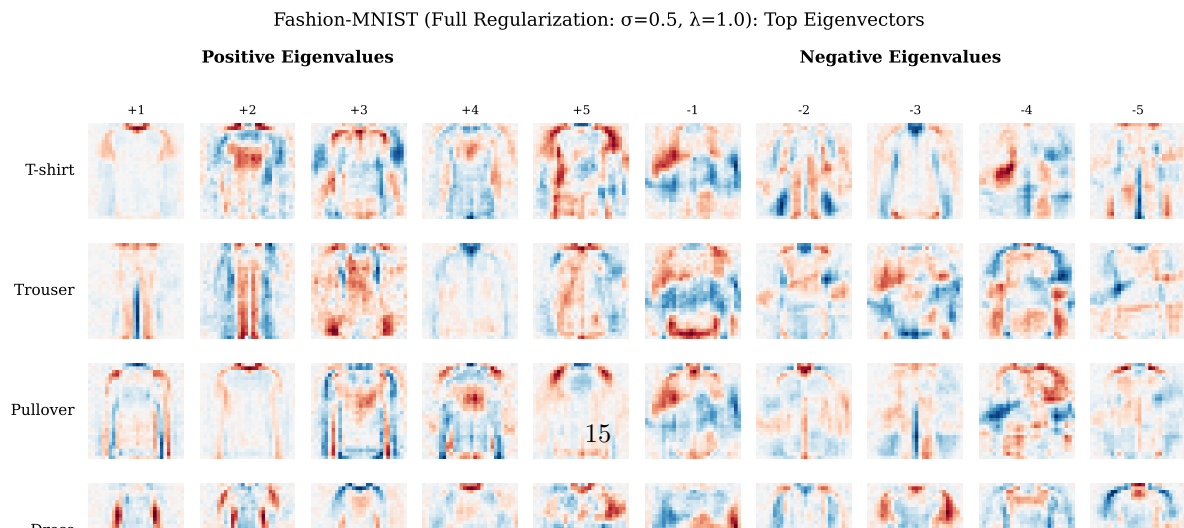


(a) No regularization: memorized outlying pixel patterns





(a) No regularization: diffuse patterns



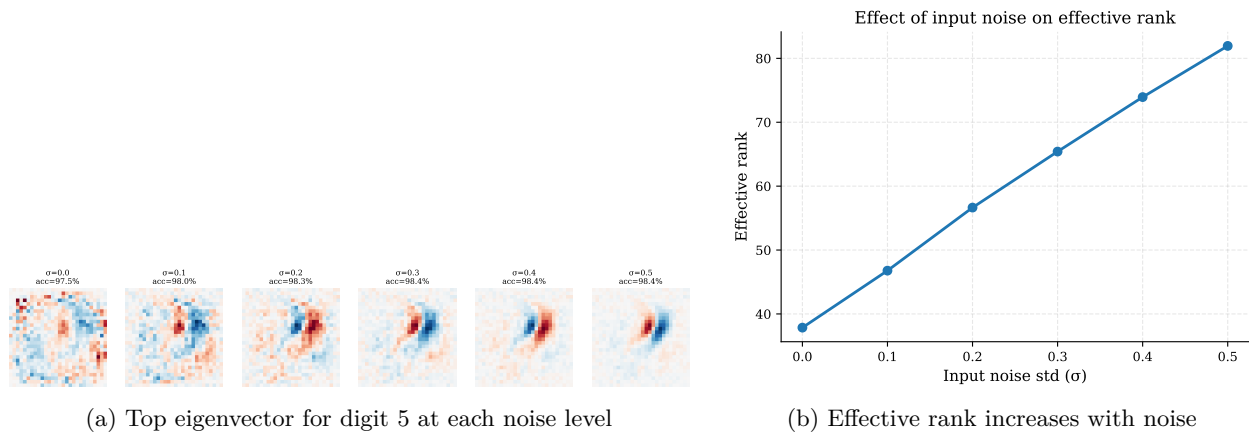


Figure 5: Effect of input noise on eigenvector interpretability (reproducing paper Figure 4). (a) As noise increases, the top eigenvector transitions from scattered pixel patterns to a recognizable digit shape. (b) Effective rank increases monotonically with noise, indicating that noise encourages distributed representations. Accuracy also improves with noise (from 97.5% at 0.0 to 98.4% at 0.5), confirming that noise regularization improves generalization.

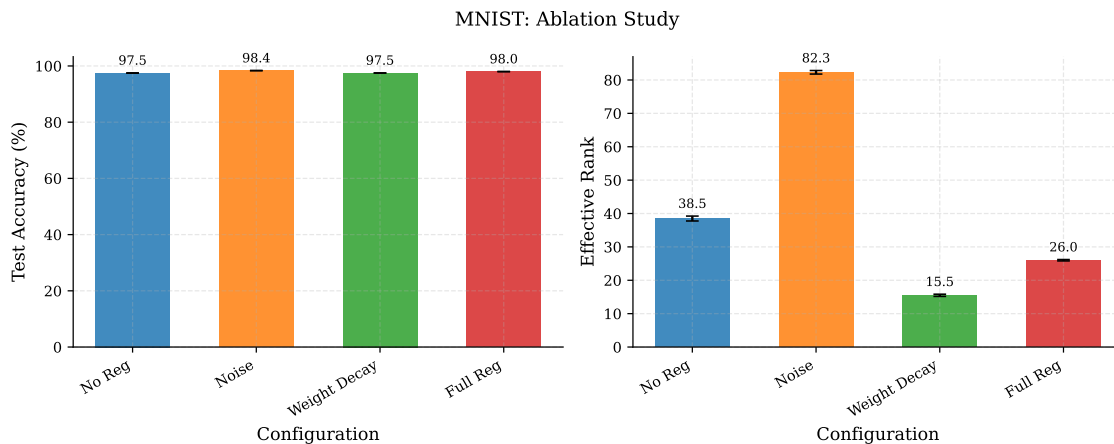


Figure 6: Ablation study visualization for MNIST showing accuracy and effective rank across regularization configurations. Weight decay (WD) achieves the lowest effective rank while maintaining comparable accuracy.

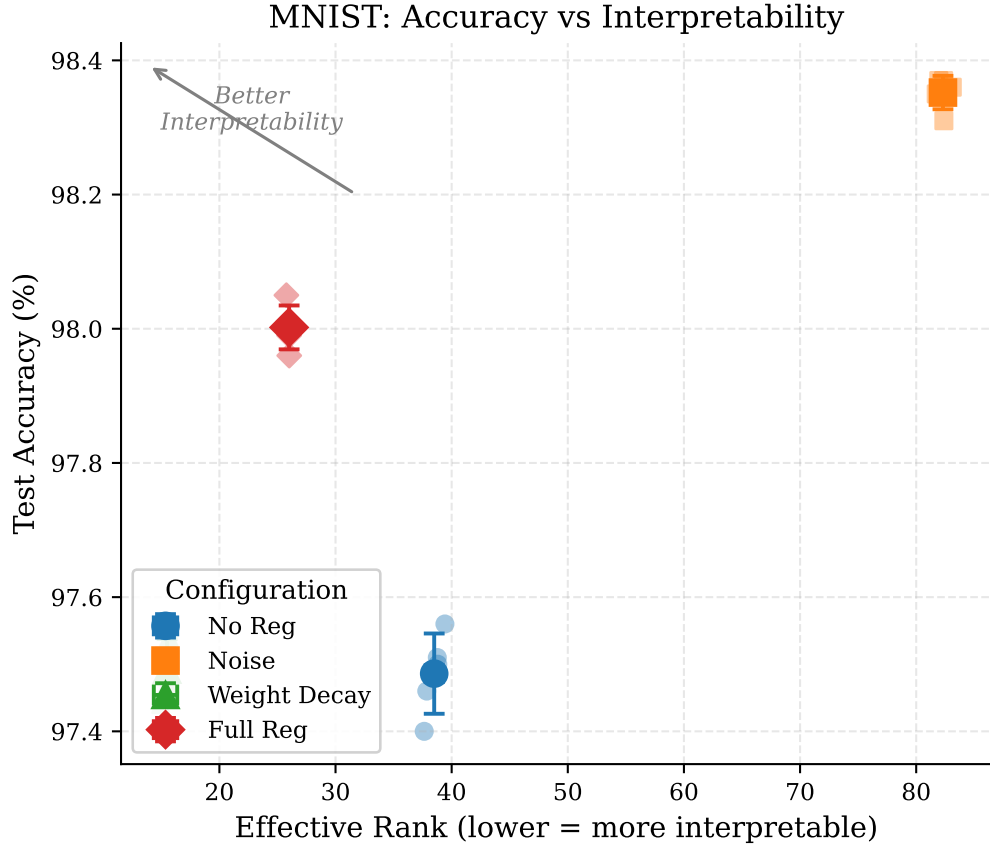


Figure 7: Accuracy vs effective rank trade-off on MNIST. Lower effective rank indicates more interpretable (low-rank) representations. Each point represents one seed; colors indicate configuration.

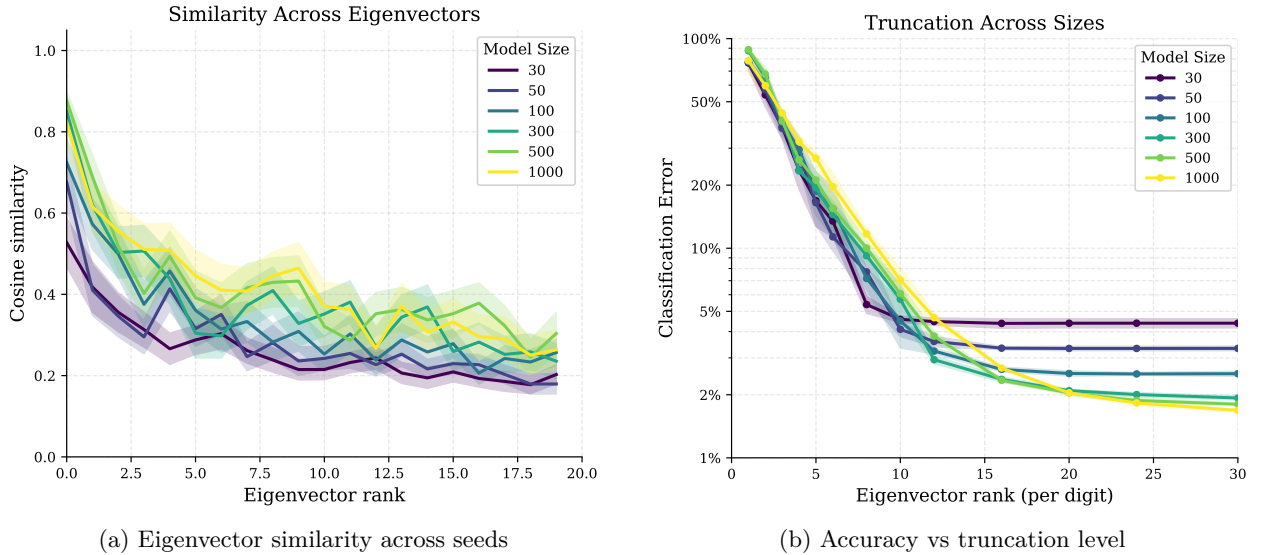


Figure 8: Model size analysis (reproducing Paper Figure 5). (a) Mean cosine similarity between top eigenvectors across different random seeds for each model size. Larger models show more consistent eigenvectors. (b) Classification accuracy when using only the top- k eigenvalues per class. Performance plateaus around $k = 10$ – 20 , confirming that a small number of eigenvectors capture most of the discriminative information.

Challenge task: eigendecomposition (True – False direction)

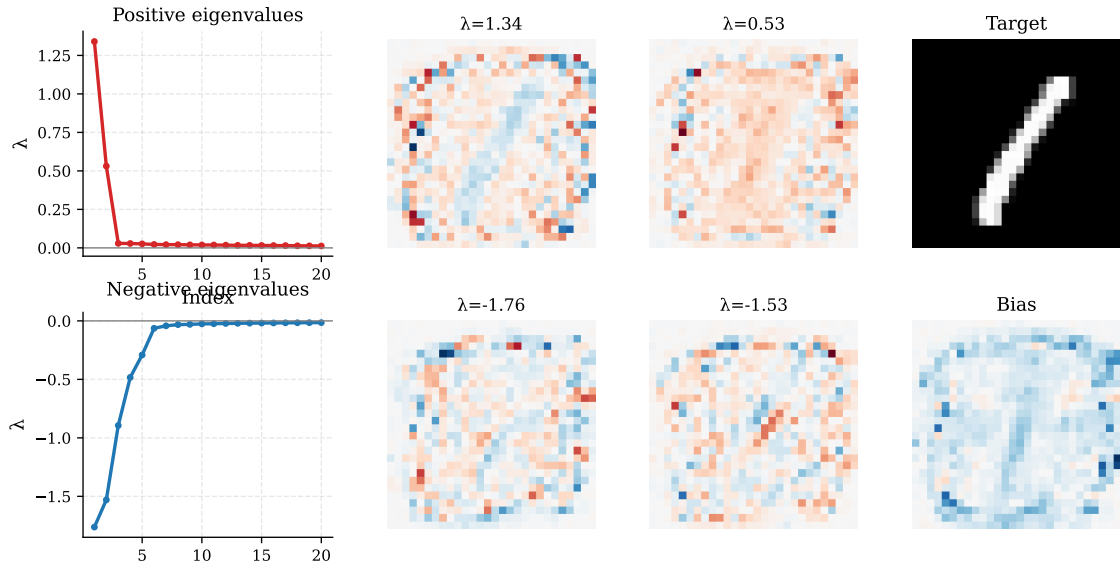


Figure 9: Challenge task eigendecomposition (reproducing Paper Figure 6). The model is trained to classify MNIST images based on similarity to a target “1” digit. Decomposition on the True–False direction reveals a dominant positive eigenvalue; the top eigenvector resembles the target digit. The learned bias encodes the similarity threshold for classification. This demonstrates that bilinear decomposition can recover the exact algorithm used for classification.

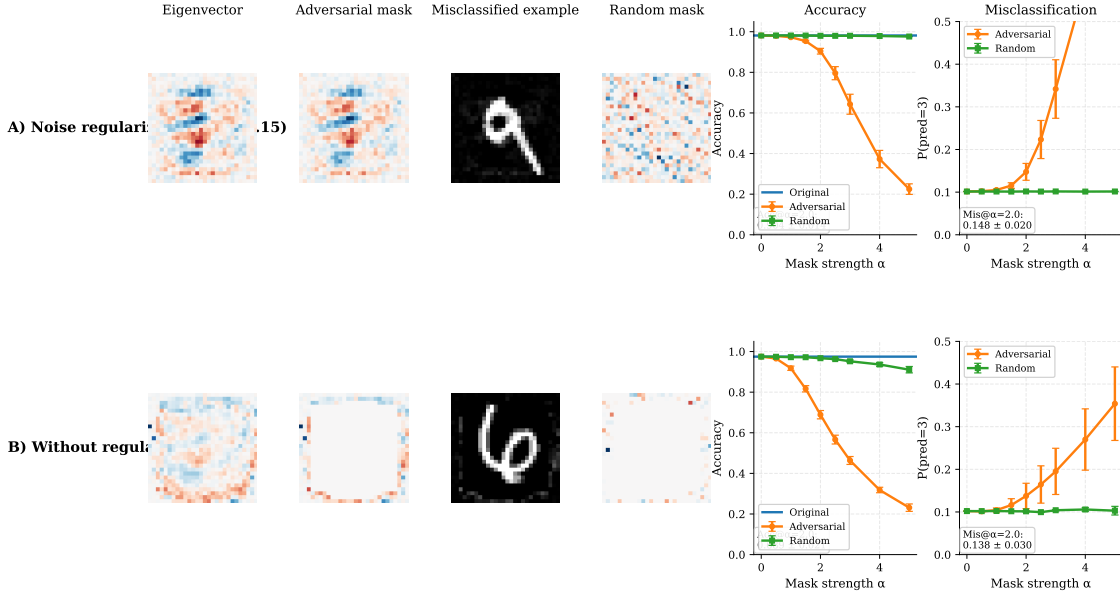
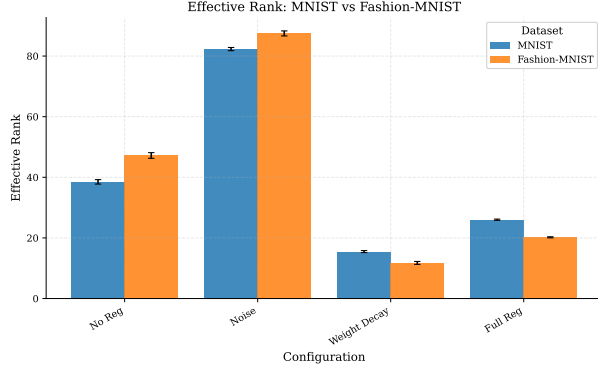
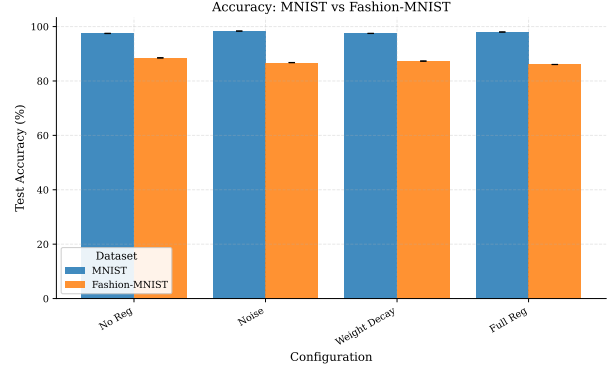


Figure 10: Adversarial mask attacks (reproducing Paper Figure 7). Top row: The adversarial mask (computed from the pseudoinverse of top eigenvectors) targets a specific digit class. Bottom row: Example images perturbed with increasing attack strength α . The accuracy plot shows that adversarial masks cause slightly more accuracy degradation than random baselines, though the effect is modest with our fully-regularized model.

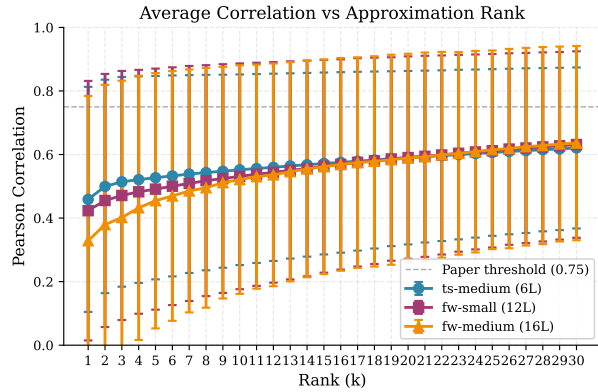


(a) Effective rank comparison

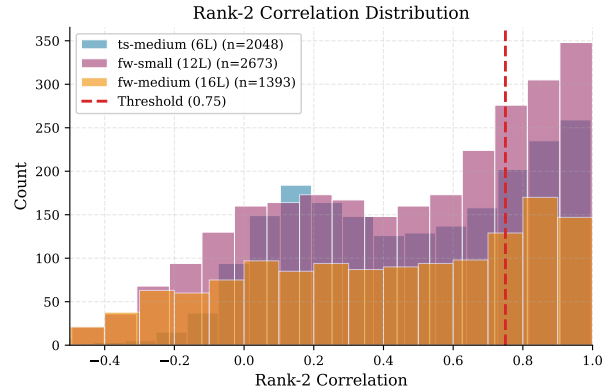


(b) Accuracy comparison

Figure 11: Cross-dataset comparison between MNIST and Fashion-MNIST. Both datasets show consistent trends across regularization configurations.



(a) Correlation vs approximation rank



(b) Rank-2 correlation distribution

Figure 12: Language model interaction matrix analysis (reproducing Paper Figure 9). (a) Mean correlation between interaction matrices and their low-rank approximations across three models. Correlation improves with rank but remains below the paper’s claimed threshold. (b) Histogram of rank-2 correlation across SAE features. Most features have low correlation with their rank-2 approximation, contrary to the paper’s claim of 69% > 0.75 (dashed line).